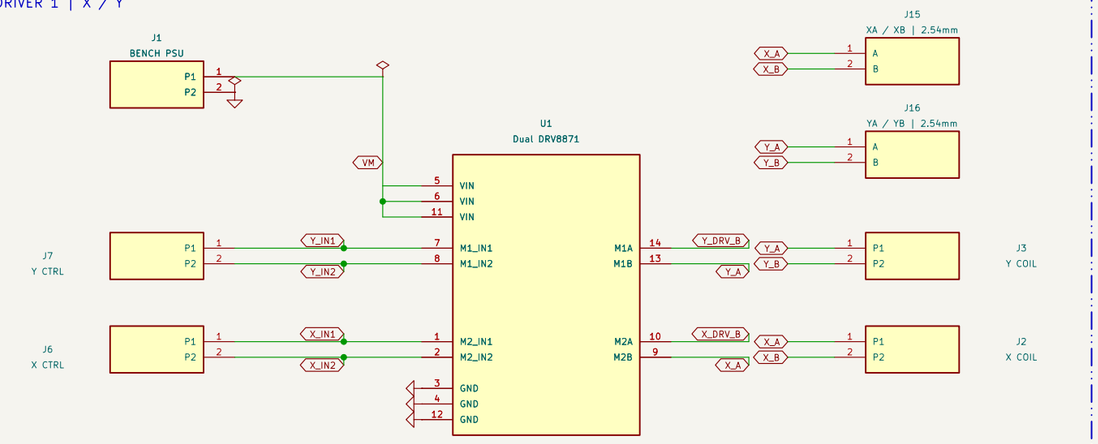
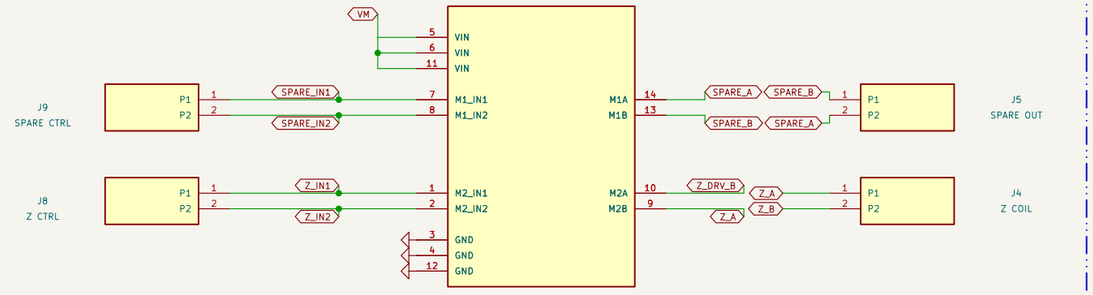
DRIVER 1 | X / Y



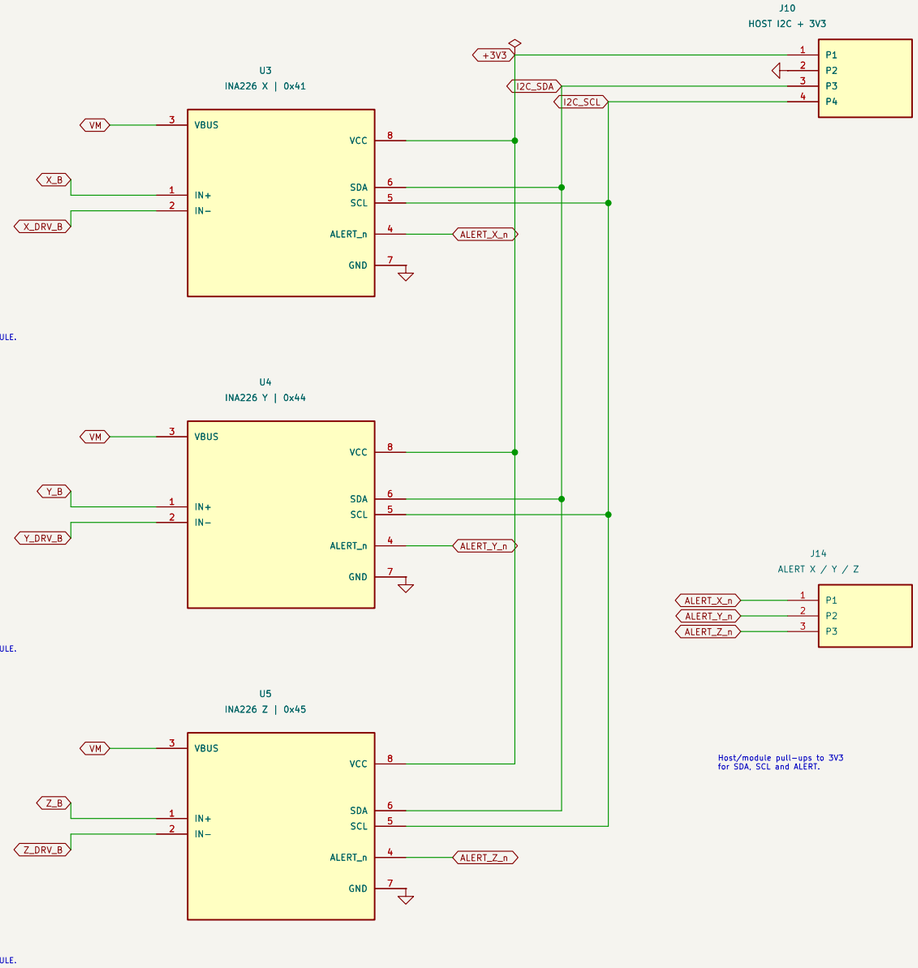
M1 = X; M2 = Y. Coil A/B are switching outputs.

DRIVER 2 | Z / SPARE



M1 = Z; M2 = SPARE. Coil A/B are switching outputs.

INA226 | SHARED I2C + 3V3 + GND



IN+ from coil B; IN− to driver B. Shunt is ON MODULE.

IN+ from coil B; IN− to driver B. Shunt is ON MODULE.

INA MODULE ADDRESS PADS (power off before soldering)
X: A0 to VCC -> 0x41; Y: A1 to VCC -> 0x44; Z: A0 and A1 to VCC -> 0x45.
Unbridged address pins must default to GND on the purchased modules. Verify settings with an I2C scan.

COIL CURRENT PATH: driver OUTA -> coil A -> coil B -> INA IN+ -> ON-MODULE SHUNT -> INA IN- -> driver OUTB.
J15/J16/J17 probe the actual coil A/B terminals, after the shunts.
PWM/flyback compatibility is not validated; INA226 inputs must remain within their specified limits.

VM = PSU positive; keep VM / VBUS at 0–36V. 3V3 comes from the host. IN+/IN- insert the module shunt in each B lead. PWM/flyback input limits remain to be validated.

MODULE FOOTPRINT PIN OFFSETS ARE PROVISIONAL: verify actual parts before fabrication. PCB placement and routing: Padam.

Sheet: /	
File: carrier_MTQ.kicad_sch	
Title: Magnetorquer test carrier	
Size: A3	Date:
KiCad E.D.A. 10.0.4	Id: 1/1

Rev: DRAFT